

No. of Functional Features Included in the Solution

Date	03 October 2026
Team ID	SWTID-2026-9455
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

This document lists all functional features included in the solution, with purpose, implementation status and contribution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Story input form	Prompt, character, setting, tone and art style with dropdowns and a loading message	templates/index.html	Done	0.5
2	Input validation	Pydantic PromptRequest; friendly error messages (422)	schemas.py, routes.py	Done	0.5
3	Outline generation	5-panel JSON outline with Gemini Flash; robust JSON cleaning and retries	services/gemini_flash.py	Done	0.5
4	Narration and dialogue	Gemini Pro expands the outline into caption and narration per panel	services/gemini_pro.py	Done	0.5
5	Panel illustrations	Stable Diffusion image per panel in the chosen art style; placeholder fallback	services/image_generator.py	Done	1.0
6	Layout builder	Matches images with story text into a structured layout	services/layout_builder.py	Done	0.5
7	PDF export	Multi-page timestamped PDF with FPDF2	services/exporters.py	Done	0.5
8	Comic preview + download + success page	Panel-by-panel preview, download button, export success page	comic_preview.html, export_success.html	Done	0.5
9	JSON API	/generate-comic/json returns layout and PDF path	routes.py	Done	0.25
10	Developer utilities	/test-image, /health, offline mock mode, interactive /docs	routes.py, mock_ai.py	Done	0.25

Marks contribution is a self-assigned split of the 5 marks (total 5.0).

Feature Summary

Metric	Count / Value
Total Features Planned	10
Total Features Implemented	10
Core / Must-Have Features	7 (features 1-7)
Additional / Nice-to-Have Features	3 (features 8-10)
Features Tested & Verified	10 (automated tests, offline mock runs, and a full end-to-end run with real Gemini and Stable Diffusion)

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Story form, comic preview, success page
2	Backend / Logic	4	Validation, layout builder, PDF export, pipeline
3	Database / Storage	1	File storage of panels and PDFs (no database)
4	API / Integration	3	Gemini Flash/Pro, Stable Diffusion, JSON API
5	Security / Authentication	1	Input validation, secrets in .env, safe download route (no login in this release)